

Design Methods for Accessibility

05332/05632/17332/17732 Fall 2026

Introduction to Accessibility and Assistive Technology

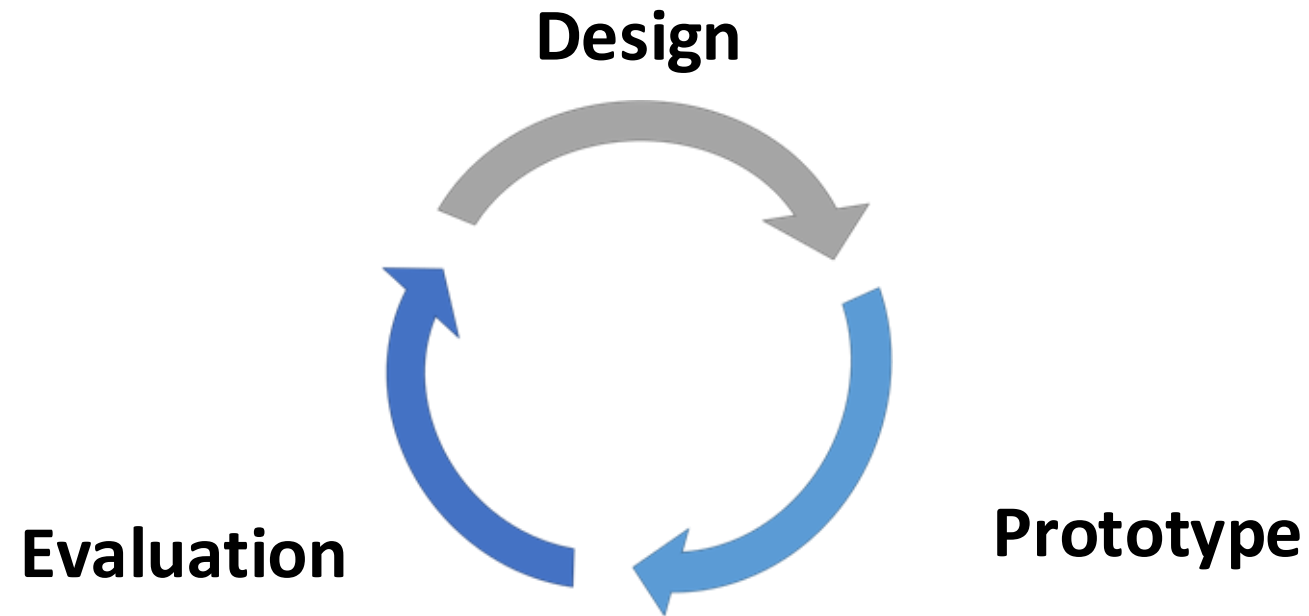
<https://cmu-intro-to-accessibility.github.io>

Andrew Begel

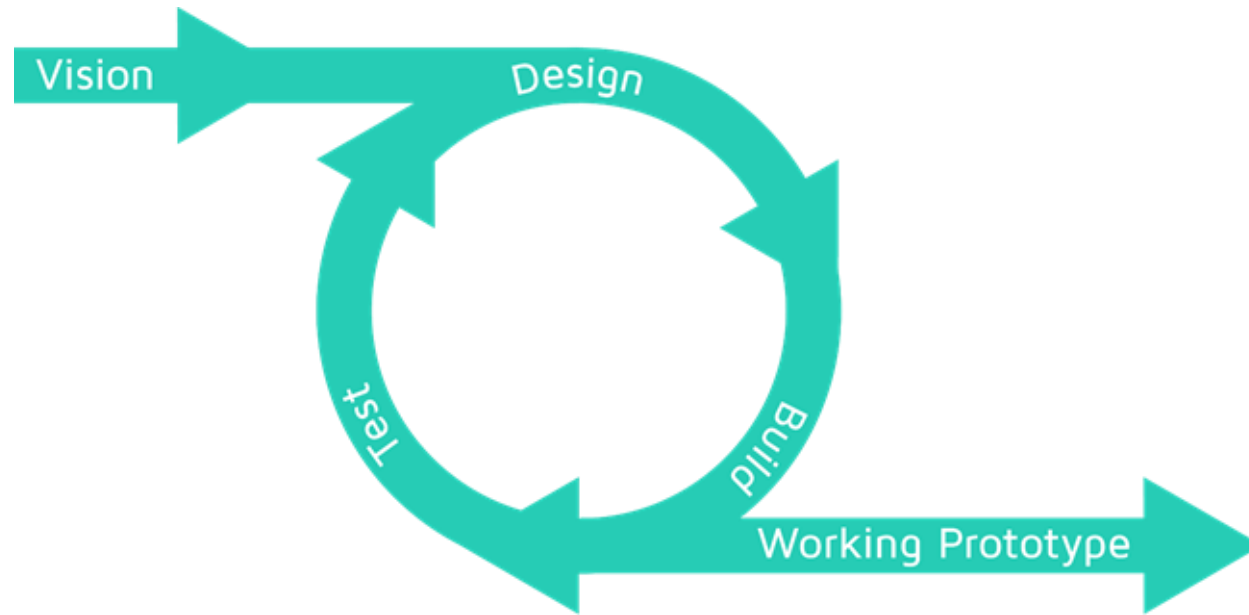
Administrivia

- HW5 due Friday October 9
- P3 due Friday October 9
- Remember your project checkin meetings this week!
- Heads up! Class field trip – Monday, November 16
 - Library of Accessible Media for Pennsylvanians (LAMP)
 - 4724 Baum Blvd. 11:30am
 - We'll walk over together from TCS Hall at 11am.

Iterative Design



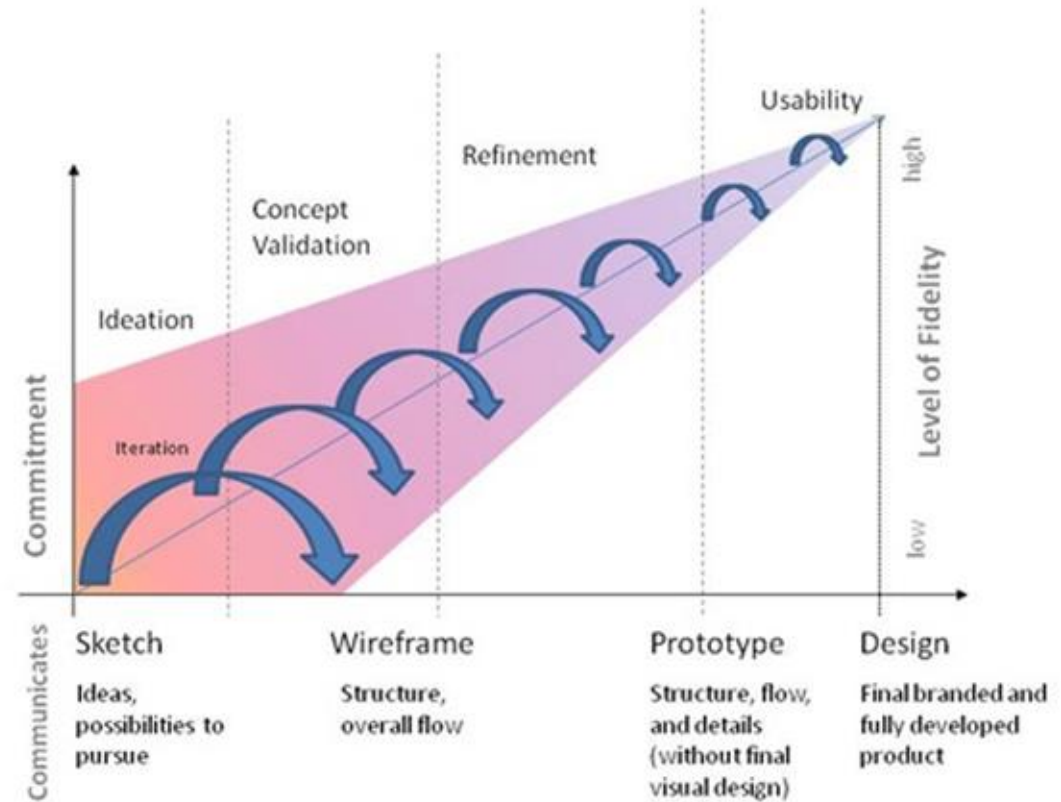
Also iterative, but with a way out



Progressive Refinement

Uncoil the loops

Increase Fidelity/Detail



Who/What are we designing for?

Oblivious Systems

Most interactive systems have
no idea about people's abilities

or the situations people are in.

What if they knew a lot more?

What could we do?



Ability Assumptions

All human
“ability”

Consider

What are

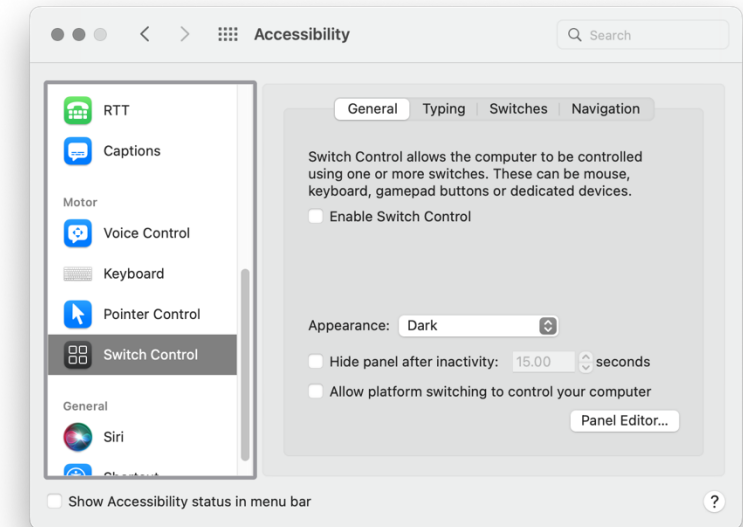
(There are



The Burden of Adaptation

Because systems are oblivious, the burden is on the user to adapt themselves to the ability/demands of interactive systems.

Operating systems offer some level of customizable adaptation to modify inputs and outputs for applications



The Burden of Adaptation

Interactive systems have no idea the user is having to do this, or why.

How can we move the burden of adaptation from the user to the system to take advantage of whatever abilities the user does have?



Assistive Tech and Rehab Engineering

Assistive Technologies are often created to enable people with disabilities to use computing systems and devices.

Rehabilitation Engineering emerges as a way to more systematically adapt to human abilities rather than trial and error approach.

Xbox Adaptive Controller



Universal Design

- A concept that involves designing products and environments to be usable by people of all abilities without the need for special accommodations. UD is based on the idea that environments should be designed to be usable by everyone without the need for adaptation.
- From architecture and in response to limitations of add-on approaches from Assistive Tech and Rehabilitation Engineering.

MACE, R. L., HARDIE, G. J., AND PLACE, J. P. 1991. Accessible environments: Toward universal design. In Design Intervention: Toward a More Humane Architecture, W. E. Preiser, J. C. Vischer, and E. T. White Eds., Van Nostrand Reinhold, New York.

Inclusive Design

- Similar to Universal Design but emphasizes the explicit inclusion of people from diverse backgrounds and abilities.
- “If you do not intentionally, deliberately and proactively include, you will unintentionally exclude.” - Joe Gerstandt

Clarkson, P. J., Coleman, R., Keates, S., & Lebbon, C. (2013). Inclusive design: Design for the whole population.



Ability-Based Design

A design approach in which the human abilities required to use a technology in a given context are scrutinized, and systems are made operable by or adaptable to alternative abilities.

(Wobbrock et al. 2011, 2014)

Comparing Design Methods

Ability-Based Design	Universal Design
• Focus on abilities of a user	• Focus on accessibility of environment
• Focus on what one person can do	• Focus on what many people can do
• Design for one	• Design for all
• Runtime adaptation	• Design-time accommodation
• Sense, model, adapt	• Understand, design, test, deploy
• Usually dynamic	• Usually fixed

Making Memes Accessible

Cole Gleason, Amy Pavel,
Xingyu Liu, Patrick Carrington,
Lydia Chilton, and Jeffrey Bigham

Carnegie
Mellon
University

meme example



meme example



audio template

top text

♪ upbeat dance music
slows to a disappointing
finish ♪

bottom text

alt-text template

Top half of penguin facing
right on a red background.
Bottom half of penguin
reversed on blue background.

Overlaid text on top:

top text

Overlaid text on bottom:

bottom text





Meme: Success Kid

Theme: small victories or little triumphs celebrated in an outsized fashion.

How do we convey the theme embedded in the meme image in alternative formats?

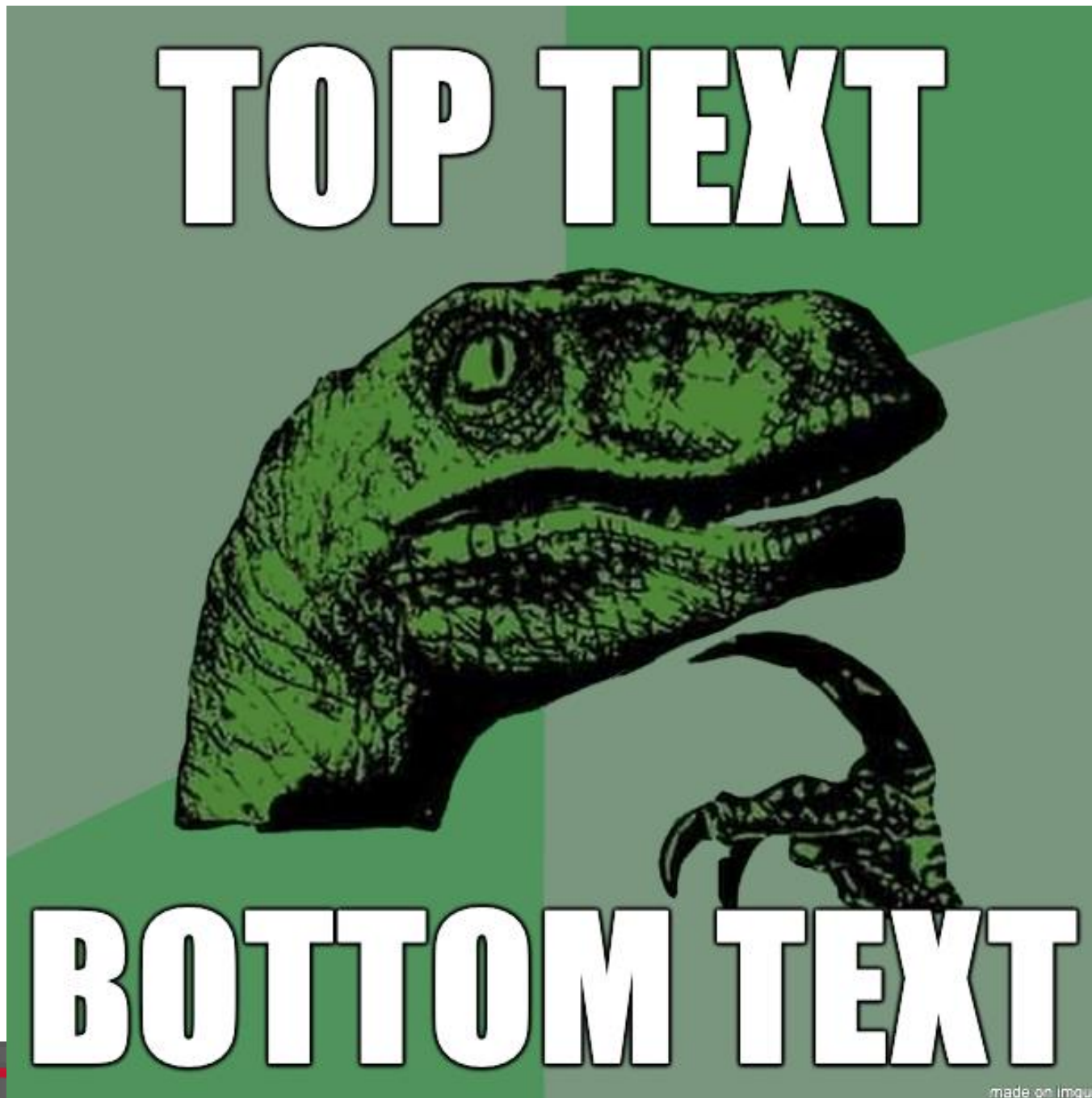
Participation Activity

- Find one of your favorite memes.
- Give the picture to an LLM and ask it to tell you what the meme is about.
- Write a post in **#in-class** with
 1. The meme
 2. The LLM interpretation of what it's about
 3. Your interpretation of what it's really about
 4. Did the LLM get it right?



Alt Text:

A drawing of a green dinosaur raptor with a claw to its chin and mouth open, as if it is contemplating something.



Alt Text:

A drawing of a green dinosaur raptor with a claw to its chin and mouth open, as if it is contemplating something. Overlaid text on top: [TOP TEXT] Overlaid text on bottom: [BOTTOM TEXT]

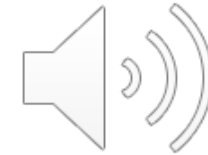


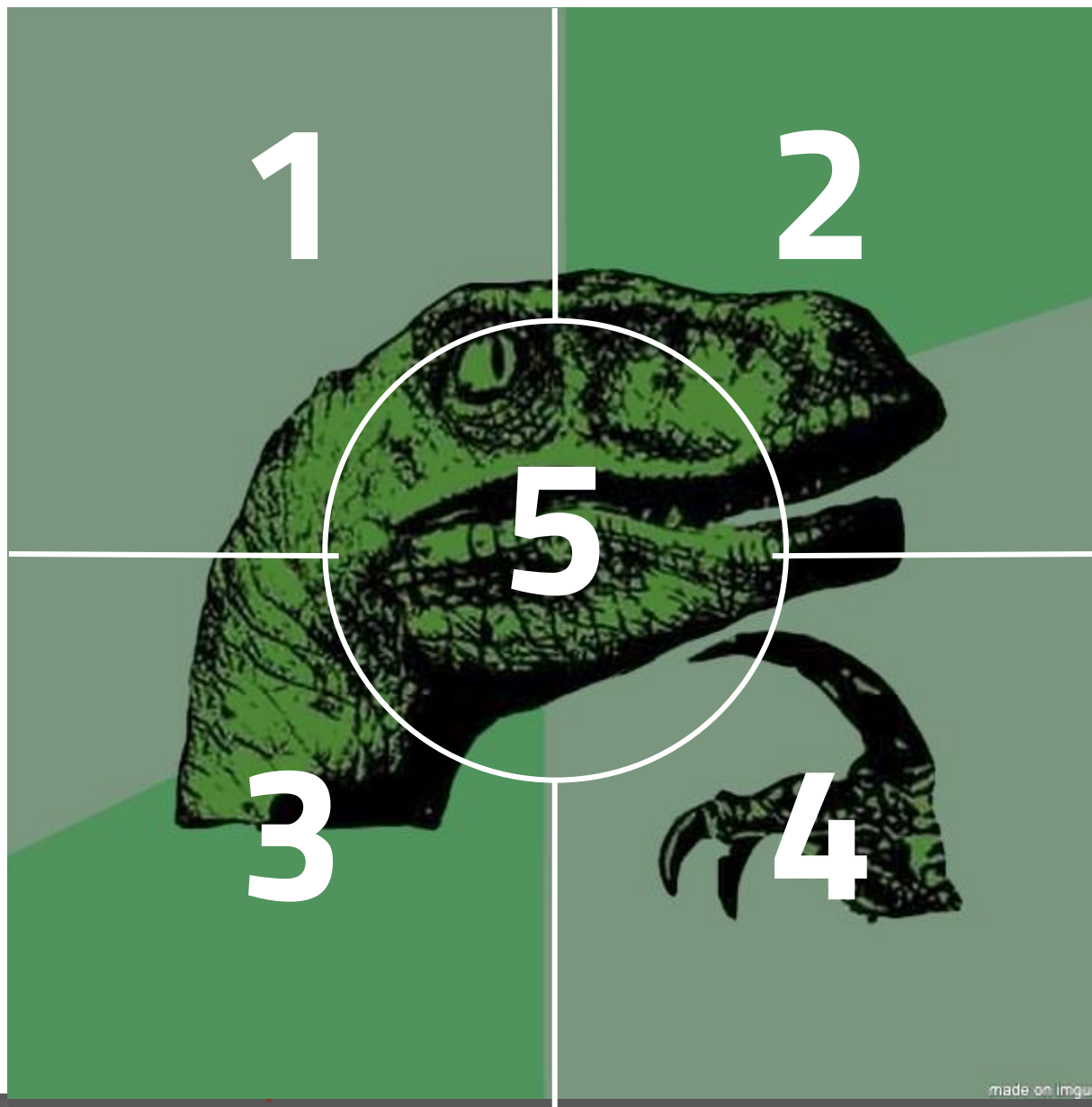
Alt Text:

A drawing of a green dinosaur raptor with a claw to its chin and mouth open, as if it is contemplating something. Overlaid text on top: WHY DO WE BAKE COOKIES Overlaid text on bottom: AND COOK BACON?



Audio template:





Recognizing the meme type

1. **Color Histograms of 5 areas**

Accuracy: 78%

1. **Structural image similarity**

Accuracy: 79%

Combined: 92% accuracy

Meme formats for User Study

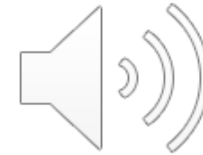
Text Only:

I DROPPED MY MACBOOK
ON MY OTHER MACBOOK

Alt Text:

Close up on a woman with her eyes closed, head in one hand, and a stream of tears running down her cheek.
Overlaid text on top: I DROPPED MY
MACBOOK Overlaid text on bottom: ON
MY OTHER MACBOOK

Audio:



“

[The alt] gives you “head in hands, crying”. I could get the emotion, but the reason for the emotion appears in the text. -- P3

It's a little confusing, because I'm like “Why is a bear saying this?” or “Why is a penguin saying this?” -- P6



Preference Results

- 8/10 participants preferred alt text memes
 - “Characters” sometimes aided understanding.
 - Uses existing infrastructure (screen reader alt)
 - More universally accessible.
- Other participants wanted more efficient methods:
 - P6: Audio quickly conveyed tone
 - P9: Text only less verbose

Participation Activity – Accessible Redesign

- Pair up with two neighbors.
- Design a new experience for one of the following:
 1. Tactile-first visual art gallery for someone with only light perception
 2. Audio-first, audio-only web browser for a motorcycle driver.
- In **#in-class**,
 1. Explain your concept that makes the experience accessible.
 2. Use an LLM to depict the experience with a 6-panel storyboard.
Post the storyboard.
 3. Tag your partners.
- 1. The most creative ideas win **PRIZES!**